Abstract – Stack-Based Browser Navigation

This project demonstrates a simplified model of web browser navigation using the Stack data structure. It simulates the fundamental browser operations such as visiting new web pages, moving backward to previously visited pages, and moving forward to pages revisited after going back.

The system is implemented using two stacks: a Back Stack and a Forward Stack. When a user visits a new page, the current page is pushed onto the Back Stack, and the Forward Stack is cleared, replicating real browser behavior. The Back operation retrieves the most recently visited page from the Back Stack and moves the current page to the Forward Stack. Similarly, the Forward operation restores pages from the Forward Stack back to the current view while updating the Back Stack.

This approach efficiently models the Last-In-First-Out (LIFO) principle of stacks, making it suitable for handling navigation history. The project helps in understanding how real-world browsers manage history internally and reinforces concepts of data structures, memory management, and algorithmic design.

The implementation is useful for educational purposes, especially in learning how stacks are applied in practical systems like web browsers.

Introduction – Stack-Based Browser Navigation

A web browser allows users to visit different web pages and navigate between them using Back and Forward buttons. Internally, this navigation is managed using a systematic approach to store and retrieve previously visited pages.

This project focuses on implementing browser navigation using the Stack data structure, which follows the Last-In-First-Out (LIFO) principle. Two stacks are used: one for back navigation and another for forward navigation. These stacks help simulate how a real browser maintains history and allows movement between pages efficiently.

When a user opens a new page, the current page is stored in the back stack, and the forward stack is cleared. When the user clicks the Back button, the current page is moved to the forward stack and the most recent page from the back stack becomes the current page. Similarly, the Forward button reverses this process.

This project helps in understanding the practical application of stacks in real-world systems and strengthens the concept of data structures through a browser simulation model.

Browser-Navigation Algorithm

Step 1: Start

Step 2: Create Back stack and Forward stack

Step 3: Open Home Page

Step 4: If new page is opened, store current page in Back stack

Step 5: Clear Forward stack

Step 6: Open new page

Step 7: If Back button is pressed, move current page to Forward stack

Step 8: Open previous page from Back stack

Step 9: If Forward button is pressed, move current page to Back stack

Step 10: Open next page from Forward stack

Step 11: Repeat steps 4 to 10 until user exits

Step 12: Stop

Browser - Navigation Flowchart

Start → Open Browser → Enter URL / Click Link → Check Cache? → Yes → Load from Cache → Display Page → End

↓

No

↓

Send HTTP Request → DNS Lookup → Connect to Server → Server Processes Request → Send HTTP Response → Render Page → Display Page → End

Methodology – Stack-Based Browser Navigation

Uses Stack Data Structure (LIFO Principle)

Components:
• Back Stack (previous pages)
• Forward Stack (forward pages)
• Current Page

Open New Page:
• Push current → Back Stack
• Clear Forward Stack
• Set new page as Current

Back Navigation:
• If Back Stack not empty:
→ Push current → Forward Stack
→ Pop Back Stack → Current
• Else: No history

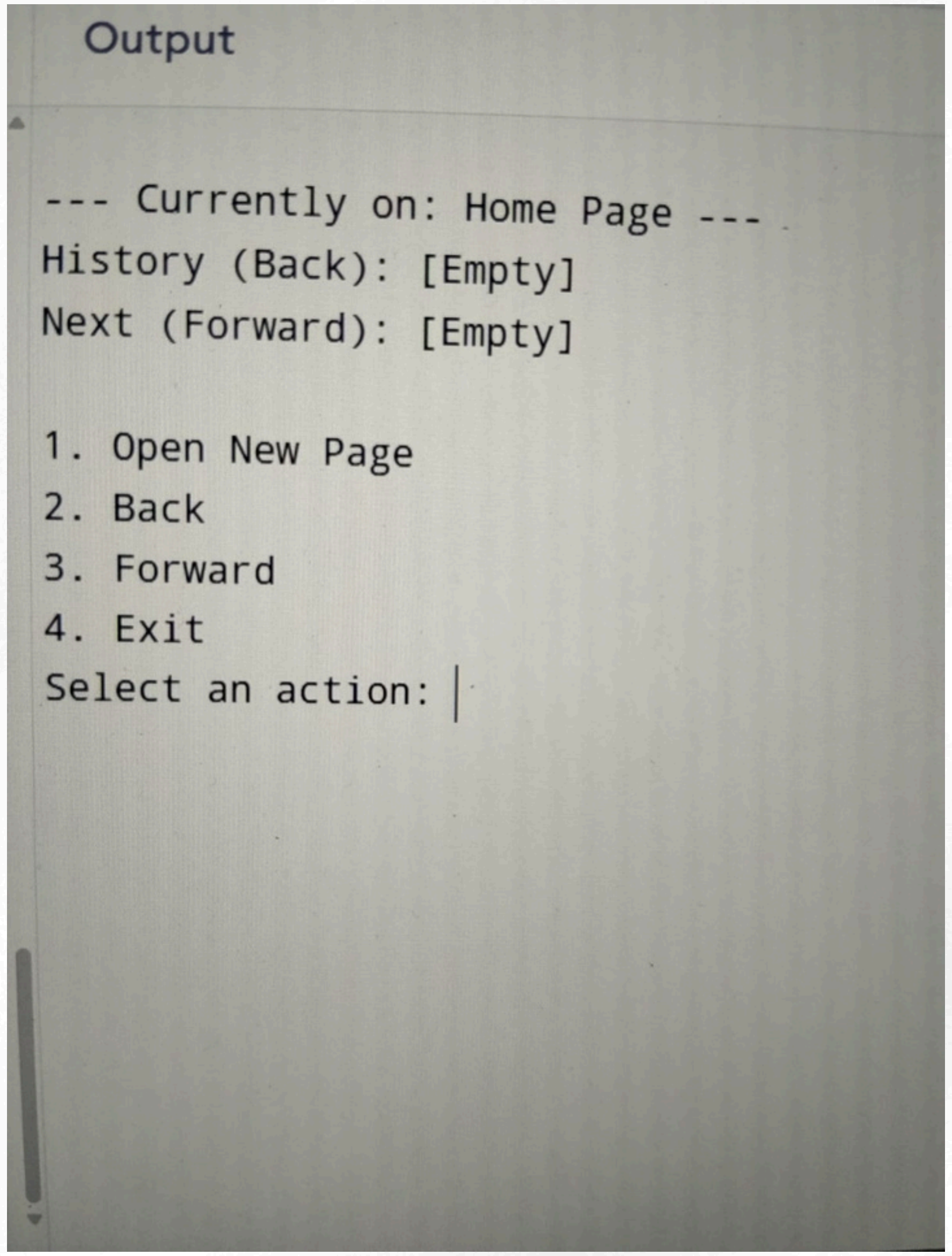
Forward Navigation:
• If Forward Stack not empty:
→ Push current → Back Stack
→ Pop Forward Stack → Current
• Else: No forward pages

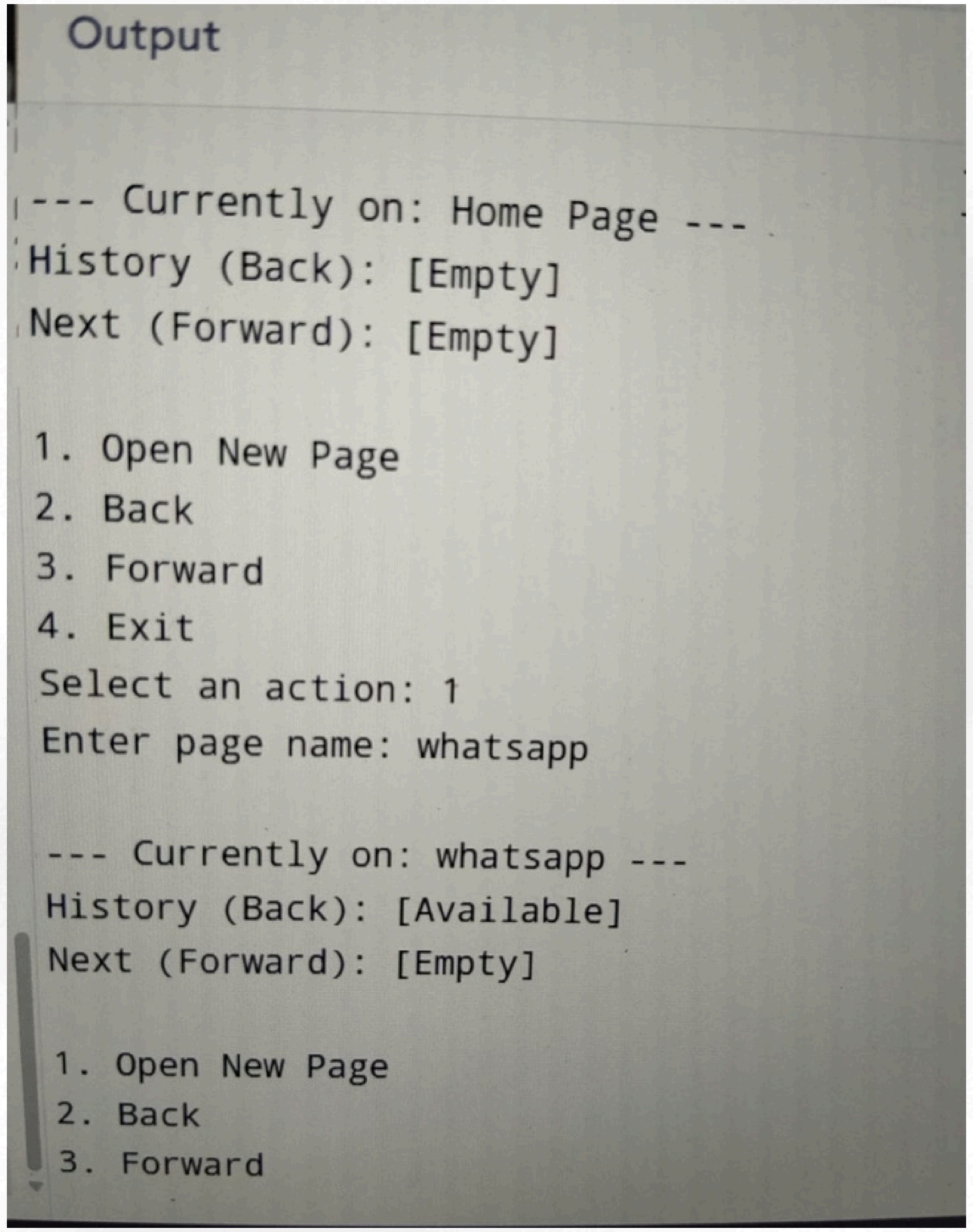
Display:
• Show current page after each action

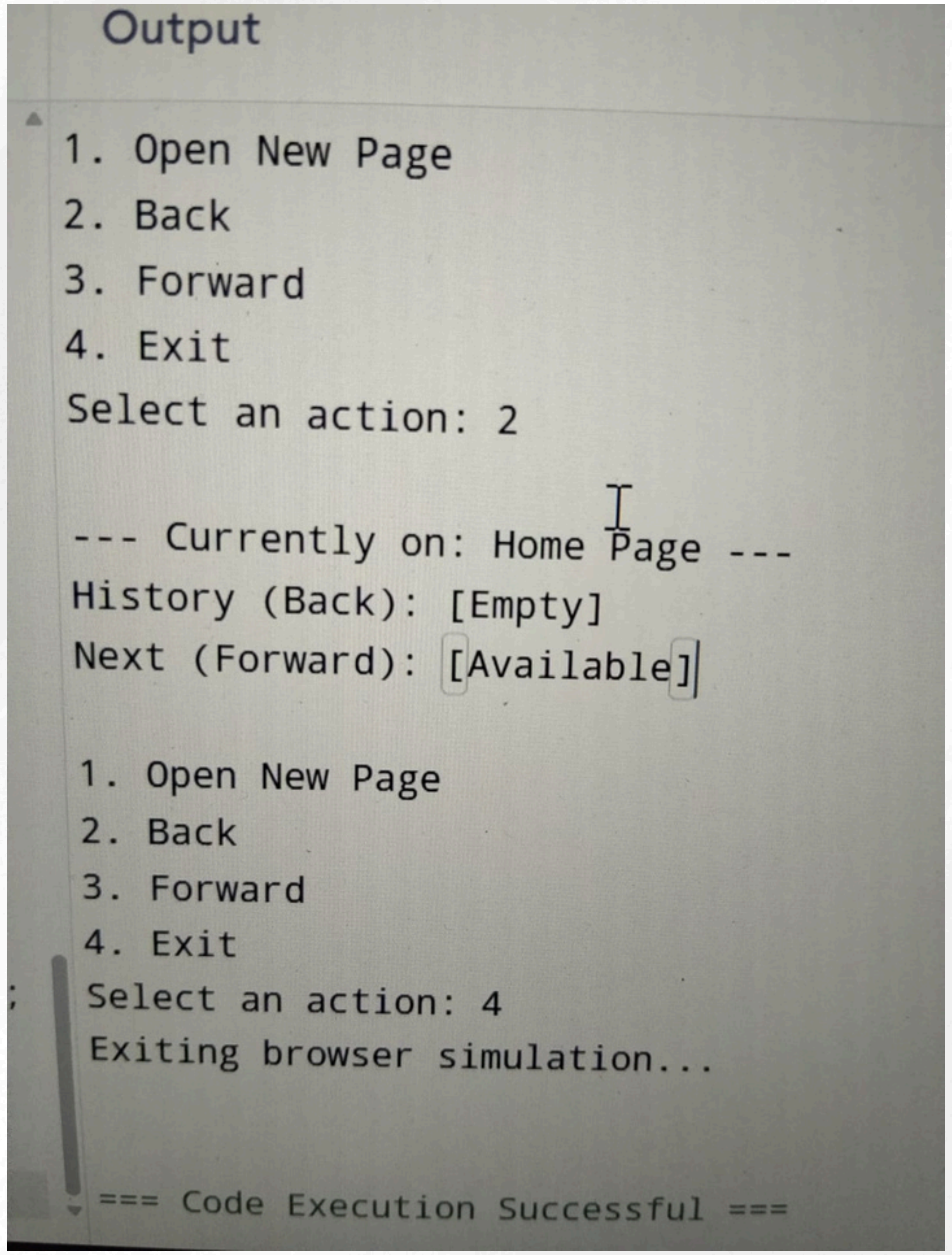
Working:
• Based on LIFO
• Uses push & pop operations

Outcome:
• Efficient navigation
• Real-world browser simulation

RESULTS


Output
--- Currently on: Home Page ---
History (Back): [Empty]
Next (Forward): [Empty]
1. Open New Page
2. Back
3. Forward
4. Exit
Select an action:


Output
--- Currently on: Home Page ---
History (Back): [Empty]
Next (Forward): [Empty]
1. Open New Page
2. Back
3. Forward
4. Exit
Select an action: 1
Enter page name: whatsapp
--- Currently on: whatsapp ---
History (Back): [Available]
Next (Forward): [Empty]
1. Open New Page
2. Back
3. Forward


Output
1. Open New Page
2. Back
3. Forward
4. Exit
Select an action: 2
--- Currently on: Home Page ---
History (Back): [Empty]
Next (Forward): [Available]
1. Open New Page
2. Back
3. Forward
4. Exit
Select an action: 4
Exiting browser simulation...
=== Code Execution Successful ===

Conclusion – Stack-Based Browser Navigation

This project successfully demonstrates the implementation of browser navigation using the Stack data structure, which follows the Last-In-First-Out (LIFO) principle. By using two stacks (Back Stack and Forward Stack), the system effectively simulates real-world browser behavior such as visiting new pages, moving backward, and moving forward through browsing history.

The project shows that stacks are highly efficient for managing navigation history because they allow easy insertion and removal of elements in a controlled order. The Back Stack stores previously visited pages, while the Forward Stack stores pages that are revisited after going back. When a new page is opened, the forward history is cleared to maintain correct browser functionality.

Through this implementation, we gain a clear understanding of how real browsers manage history internally and how fundamental data structures like stacks can be applied to real-life systems.

Overall, this project enhances practical knowledge of data structures, improves problem-solving skills, and demonstrates the real-world application of stack operations in web navigation systems.